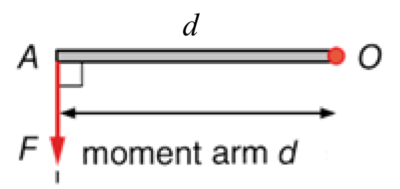
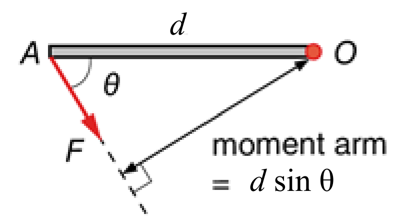
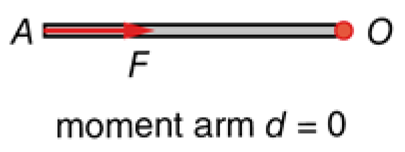
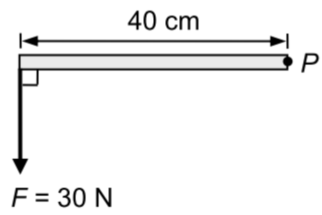
Moment of a force about a point = Force x Perpendicular distance from the point

- Moment of a force about a point is the **turning effect** of the force **about that point**.
- Moment of a force can be called **Torque**.
- The direction of moment of a force can be **clockwise** or **anticlockwise**.
- **Unit** of moment: **Nm**

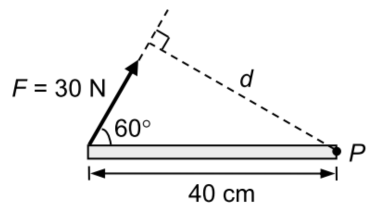
		
Moment about O $= Fd$	Moment about O $= Fd \sin\theta$	Moment about O $= 0$

1. In each case below, find the moment of force F about point P.

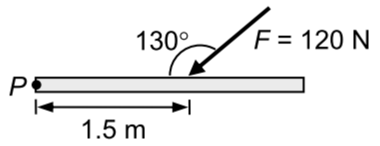
(a)



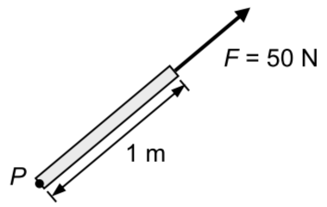
(b)



(c)



(d)



Apply $\tau = Fd_{\perp} = Fd \sin\theta$.

(a) $\tau = 30 \times 0.4$
$= 12$ N m (anticlockwise)

(b) $\tau = 30 \times 0.4 \sin 60°$
$= 10.4$ N m (clockwise)

(c) $\tau = 120 \times 1.5 \cos 40°$
$= 138$ N m (clockwise)

(d) $\tau = 50 \times 1 \sin 0°$
$= 0$

2. Find the torque acting on the door when

(a) $\theta = 30°$

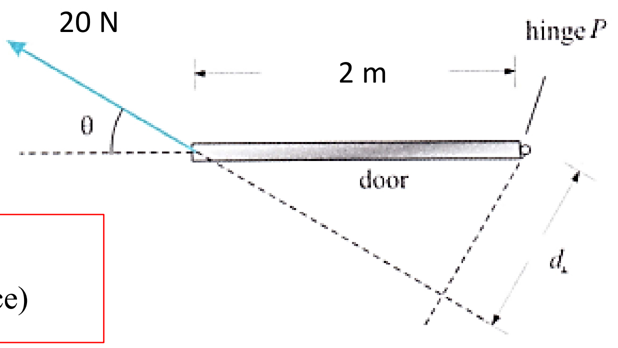
$20 \sin 30° (2) = 20Nm$

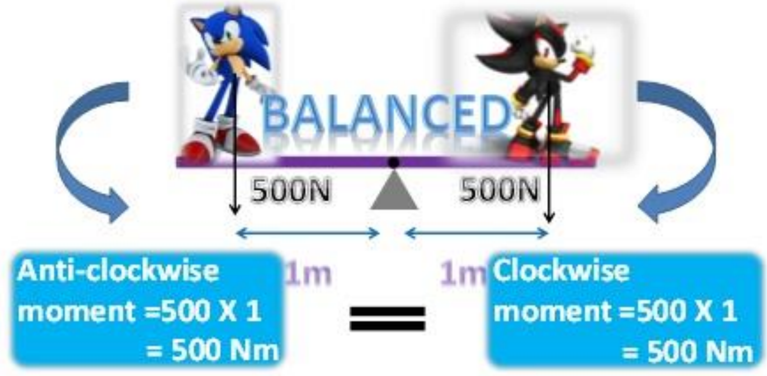
(b) $\theta = 45°$

$20 \sin 45° \ (2) = 28.3Nm$

(c) $\theta = 0°$

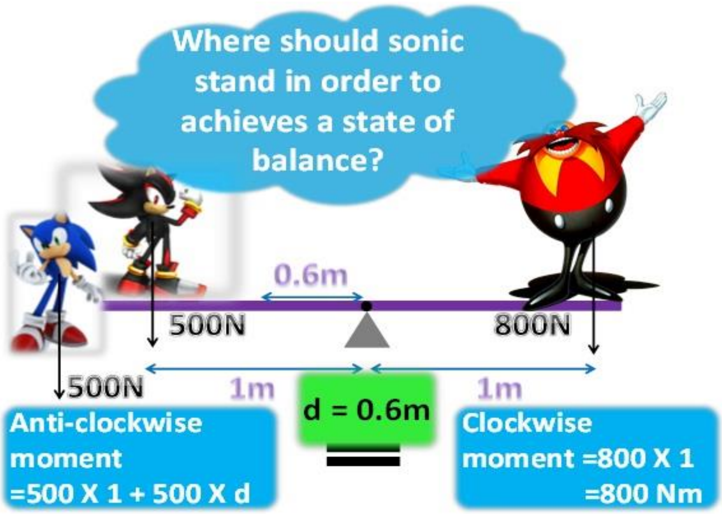
0 Nm
(because no perpendicular distance)





Principle of moments

An object achieves a state of balance when the **sum of the clockwise moments** is **equal** to the **sum of the anti-clockwise moments**.

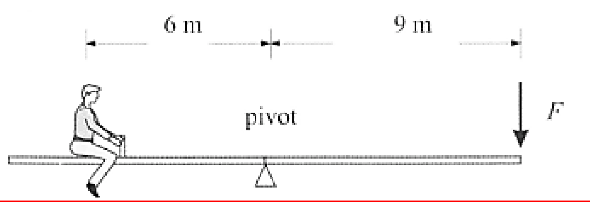


3. A seesaw is initially balanced. A boy weighted 600 N sits 6 m away from the pivot of the seesaw as shown. What force F, 9 m from the pivot, is needed to balance the seesaw?

A. 300 N
B. 400 N
C. 450 N
D. 600 N

Taking moment about the pivot,
$600 \times 6 = F \times 9$
$\therefore \quad F = 400$ N

Answer is B.



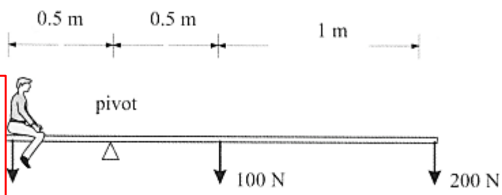
F

B

4. In the diagram, if the wooden plank of weight 100 N is in balanced, the weight of the boy is

A 350 N
B 500 N
C 700 N
D 1400 N

Let W be the weight of the boy. Taking moment about the pivot,
$W \times 0.5 = 100 \times 0.5 + 200 \times 1.5$
$\therefore \quad W = 700$ N



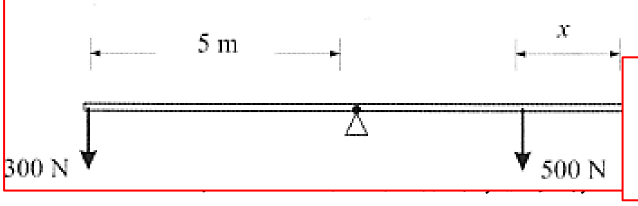
C

5. Two boys of weight 500 N and 300 N are sitting at both ends of a seesaw 10 m in length. The seesaw without any load is initially at rest. How far must the heavier boy move in order to balance the seesaw?

A. 5 m
B. 4 m
C. 3 m
D. 2 m

D

Answer is D.

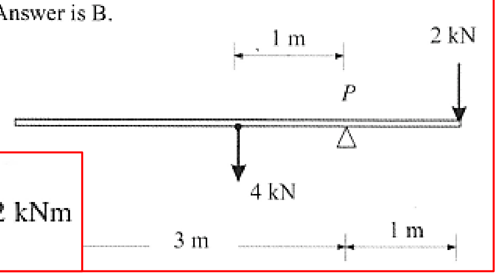


Let x be the distance moved by the heavier boy. Taking moment about the pivot,
$500 \times (5 - x) = 300 \times 5$
$\therefore \quad x = 2$ m

6. In the diagram, a uniform beam of weight 4 kN is pivoted at P and a force of 2 kN is exerted at one end. The resultant moment about P is

A. 2 kN m clockwise
B. 2 kN m anti-clockwise
C. 6 kN m clockwise
D. 6 kN m

Answer is B.



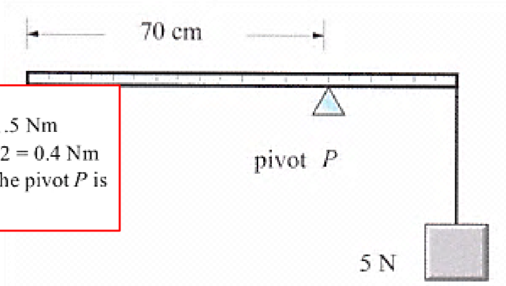
Taking moment about P,
Resultant moment $= 4 \times 1 - 2 \times 1 = 2$ kNm
anti-clockwise

B

7. A uniform meter rule of weight 2 N is pivoted at the 70 cm mark as shown. A mass of weight 5 N is suspended at the 100 cm end. When the rule is horizontal, what is the resultant turning moment about the pivot P?

A. 0.5 Nm clockwise
B. 0.5 Nm anti – clockwise
C. 1.1 Nm clockwise
D. 1.1 Nm anti – clockwise

Take moment about P.
Clockwise moment $= 5 \times 0.3 = 1.5$ Nm
Anti-clockwise moment $= 2 \times 0.2 = 0.4$ Nm
The resultant turning moment about the pivot P is $1.5 - 0.4 = 1.1$ Nm clockwise.

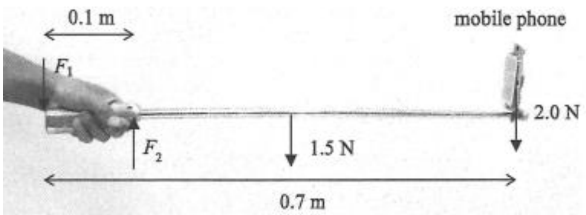


C

8. Selfie sticks are popular nowadays. A uniform selfie stick of length 0.7 m is held horizontally as shown. Assume that the forces required to hold the selfie stick by the hand are represented by F_1 and F_2, and F_1 and F_2 are perpendicular to the stick.
It is given that the weight of the selfie stick and the mobile phone are 1.5 N and 2.0 N respectively. Taking the mobile phone as a point mass, estimate the magnitude of F_2.

A. 3.5 N
B. 19.3 N
C. 35 N
D. Cannot be determined as F_1 is unknown

$(1.5)(0.35) + (2.0)(0.7) = F_2(0.1)$
$F_2 = 19.3$ N

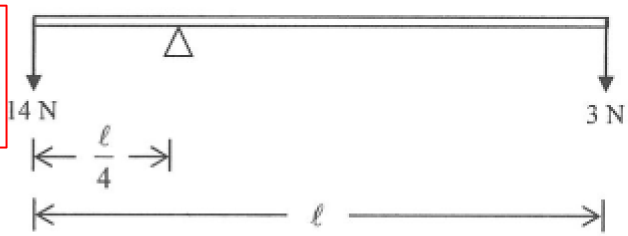


B

9. A uniform of length l is pivoted at $\frac{1}{4}\,l$ from one of its ends. Two forces 14 N and 3 N act on its two ends as shown above. If one rod is in equilibrium, find the weight of the rod.

A. 2.5 N
B. 5 N
C. 8 N
D. 11 N

Take moment about the pivot,
$14 \times \frac{\ell}{4} = W \times \frac{\ell}{4} + 3 \times \frac{3\ell}{4} \qquad \therefore\ W = 5\text{ N}$

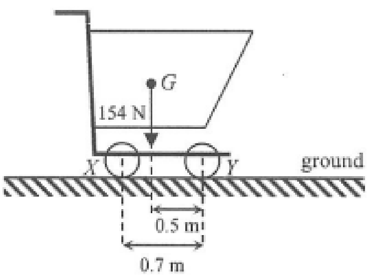


B

10. The figure shows a supermarket trolley resting on the ground. The separation between cylindrical wheels X and Y is 0.7 m. When the trolley is loaded to a total weight of 154 N, its centre of gravity G is at a horizontal distance of 0.5 m from the wheel Y. What is the reaction acting on the wheel X from the ground?

A. 44 N
B. 62 N
C. 92 N
D. 110 N

D
Take moment about Y.
$R_X \times (0.7) = (154) \times (0.5) \qquad \therefore\ R_X = 110$ N



D

Conditions for equilibrium

1. **No net force**
   (No translational motion)
2. **No net moment** (Sum of clockwise moment = Sum of anticlockwise moment)
   (No rotational motion)

Take moment about *B*,

Clockwise moment = anticlockwise moment

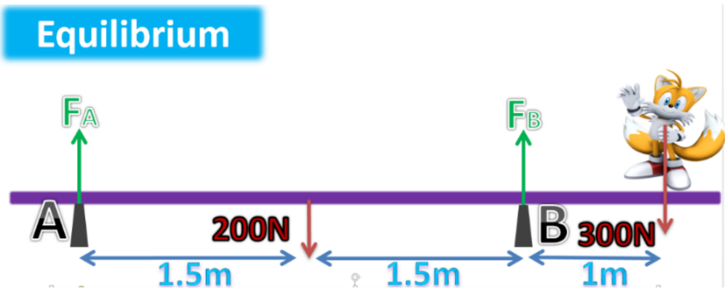
$300 \times 1 + F_A \times 3 = 200 \times 1.5$

$F_A = 0$ N

At equilibrium, net force = 0 N

$F_A + F_B = 500$ N

$F_B = 500$ N



11. The diagram shows a uniform beam XY of length 1 m supported at P where PX = 20 cm. The beam weighs 10 N. The beam is balanced by an applied force F exerted at X. (a) What is the value of F? (b) What is the reaction at P?

Va

A. 1

B. 1

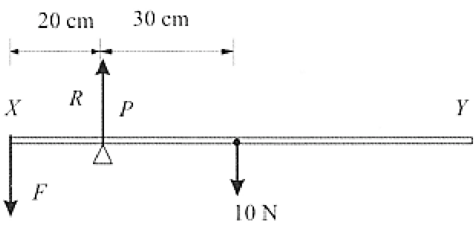
C. 2

D. 2

As the beam is uniform, the C.G. is at 30 cm from P.
Taking moment about P,
$F \times 20 = 10 \times 30$
$\therefore \quad F = 15$ N

Vertical forces are balanced:
$R = F + 10 = 15 + 10$
$= 25$ N

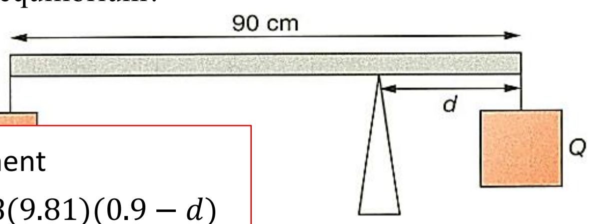
Answer is B.



B

12. Two objects P and Q are hung from a uniform bar of mass 0.4 kg as shown. The masses of P and Q are 0.8 kg and 3.2 kg respectively. The distance between the string holding Q and the pivot is d and the length of the bar is 90 cm. What is d if the bar is in equilibrium?

A. 13.5 cm
B. 18 cm
C. 20.5 cm
D. 22.5 cm



Clockwise moment=anti-clockwise moment

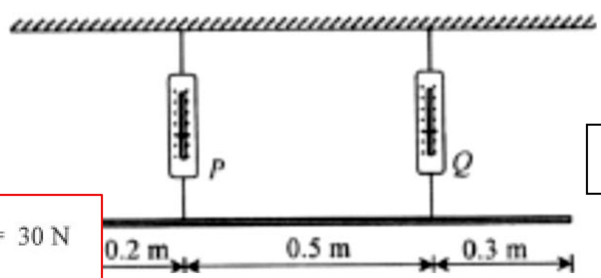
$3.2(9.81)d = 0.4(9.81)(4.5 - d) + 0.8(9.81)(0.9 - d)$

$d = 0.205m$

C

13. A uniform rod of weight 50 N is supported by two spring balances *P* and *Q* and remains at rest as shown above. Assume the weight of the rod acts through its mid-point. Find the readings of *P* and *Q*

	Reading of *P*	**Reading of *Q***
A.	17N	33N
B.	20N	30N
C.	30N	20N
D.	33N	17N



B

Take moment at the point under balance P : $50 \times 0.3 = Q \times 0.5 \quad \therefore Q = 30$ N

Balance of forces : $P + 30 = 50 \quad \therefore P = 20$ N

14. Figure (a) shows a uniform plank supported by two spring balances P and Q. The readings of the two balances are both 150 N. P is now moved 0.25 m towards Q (see Figure (b)). Find the new readings of P and Q.

C

	Reading of P / N	Reading of Q / N
A.	100 N	200 N
B.	150 N	150 N
C.	200 N	100 N
D.	200 N	150 N

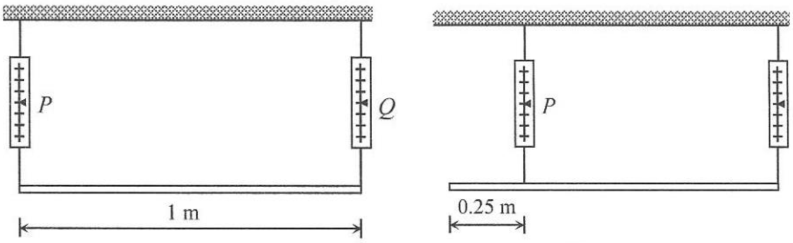


Figure (a) Figure (b)

Weight of the plank $= 150 + 150 = 300$ N

In Figure (b), take moment about the point under balance Q :

$P \times (1 - 0.25) = 300 \times 0.5$ $\quad\therefore P = 200$ N

Balance of forces : $200 + Q = 300$ $\quad\therefore Q = 100$ N

15. A plank of negligible mass is supported by two trestles. Two forces act on the two ends of the plank, and the ratio of the two forces is 2 : 3. Let F_1 and F_2 be the force exerted on the left and right trestles respectively. What is the ratio $F_1 : F_2$?

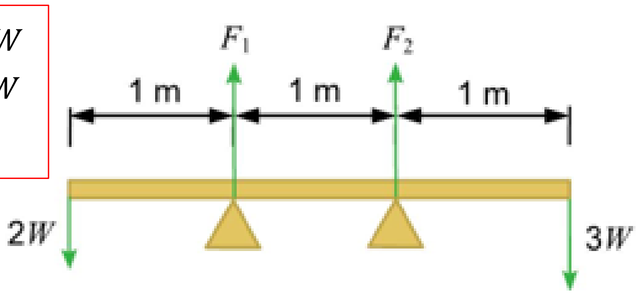
B

A. 1 : 1
B. 1 : 4
C. 4 : 1
D. 1 : 3

Take F_1 as pivot, $F_2 + 2W = 3W \times 2, F_2 = 4W$

Take F_2 as pivot, $F_1 + 3W = 2W \times 2, F_1 = W$

$\therefore F1 : F2 = 1:4$



16. A uniform plank XY of weight 150 N is support by two trestles at P and Q as shown. A girl of weight 400 N stands on the plank and the plank remains in equilibrium. What is the minimum distance between the girl and X before the plank loses balance?

B

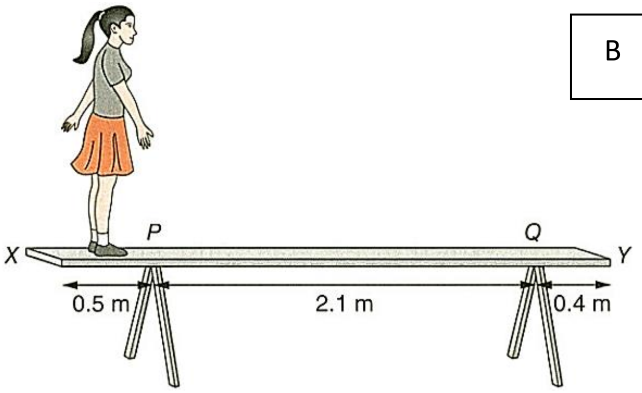
A. 0.106 m
B. 0.125 m
C. 0.245 m
D. 0.375 m

Let the distance between the girl and X be d.
Take moment about point P.
When the plank is about to lose balance, the reaction force provided by the trestle at Q is 0.

Clockwise torque = anticlockwise torque

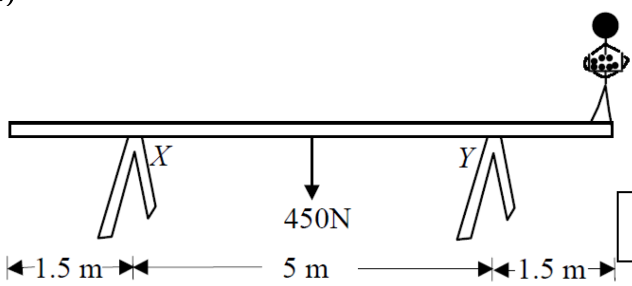
$$150\left(\frac{0.5 + 2.1 + 0.4}{2} - 0.5\right) = 400(0.5 - d)$$

$$d = 0.125 \text{ m}$$



17. A uniform plank of weight 450 N rests on two trestles X and Y and a worker of weight 675 N stands at one end of the plank as shown above. The worker holds a light basket which contains several packets of goods each of weight 6 N. What is the maximum number of packets he can hold without tilting the plank? (Assume the weight of the plank acts through its centre.)

A. 11
B. 12
C. 13
D. 18



B

Let the maximum number of packets he can hold be n.
When he holds the maximum packets, the plank will just tilt so that X will lose contact with the plank.
Take moment about Y,
$450 \times 2.5 = (675 + 6n) \times 1.5$ $\quad\therefore n = 12.5$
$\therefore$ The maximum number of packets is 12 without tilting the plank.

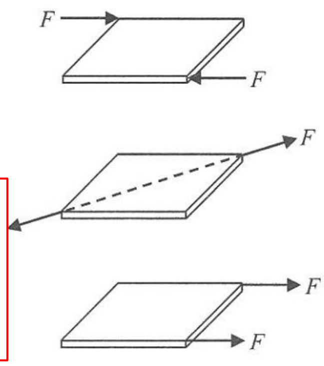
18. A block is initially at rest on smooth horizontal ground. Two forces of equal magnitude F act on the block. In which of the below cases will the block remain at rest ?

A. (2) only
B. (3) only
C. (1) & (2) only
D. (1) & (3) only

A

(1)

(2)



× (1) The two forces form a couple and cause the block to rotate, thus the block will not remain at rest.

✓ (2) The two forces will balance each other, so there is no net force acting on the block, the block will remain at rest.

× (3) The two forces give a resultant force towards the right causing an acceleration towards the right, thus the block will not remain at rest.

19. A rigid rod PQ is hinged smoothly to a wall at one end while the other end is connected by an inextensible light string to the wall at point R as shown in the figure. A weight W is suspended from a point on the rod remains horizontal. Which of the following changes would increase the tension in the string?

(1) Shift W towards the P end.
(2) Use a heavier rod
(3) Use a shorter string

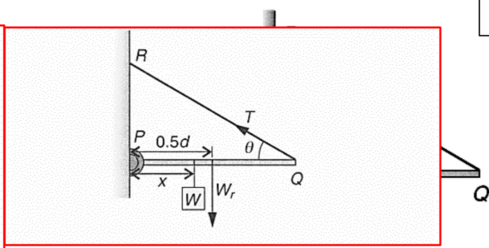
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

D

Take moment about P,
$Td\sin\theta = Wx + W_r(0.5d)$
where W_r is the weight of the rod.

$\therefore\quad T = \dfrac{Wx + 0.5W_r d}{d\sin\theta}$

(1) Shifting W towards the P end will decrease x, Hence T will be decreased.
(2) Using a heavier rod will increase W_r, Hence T will be increased.
(3) Using a shorter string will decrease θ and so as $\sin\theta$, Hence T will be increased.



20. A uniform light rigid rod OX is hinged smoothly to a wall at one end O. Its mid-point M is connected by a light inextensible string to a point Y directly above O while a weight W is suspended from the other end X of the rod as shown. Rod OX remains horizontal. The reaction force acting on the rod due to the wall is along the direction

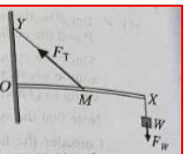
A. OP
B. OQ
C. OR
D. OS

C

C

Method 1

Let F_W be the weight of W;
F_T be the tension acting on the rod by the string; and
F_R be the reaction force acting on the rod due to the wall.

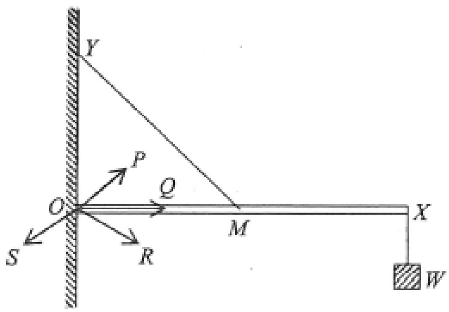


Note that the rod is at equilibrium.

Take moment about M. Since F_T does not produce any moment and F_W produces a clockwise moment, F_R produces an anticlockwise moment. Hence, the vertical component of F_R points downwards.

Take moment about Y. Since F_T does not produce any moment and F_W produces a clockwise moment, F_R produces an anticlockwise moment. Hence, the horizontal component of F_R points rightwards.

Note: In the above work, the weight of the rod is neglected.



21. A light rod OX is hinged smoothly to a wall at one end O. A weight of 50 N is attached to the other end X of the rod. The rod is held horizontally by a rope attached to its mid-point M as shown. The angle between the rod and the rope is 30°. Find the magnitude of the reaction force acting on the rod by the wall at O.

A 0
B 76.1 N
C 173 N
D 180 N

D

Let the distance from O to M be a.

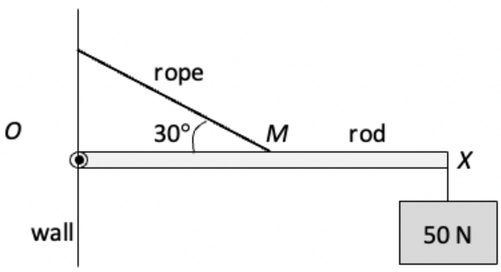
Take moment by O,

$$T\sin 30° a = 50(2a)$$

$$T = 200N$$

$$\begin{cases} R\sin\theta = T\cos 30° \\ R\cos\theta = T\sin 30° - 50 \end{cases}$$

$$\theta = 73.9°, R = 180°$$



22. A uniform gangplank of a ferry smoothly hinged at end P initially rests horizontally on the pier. The gangplank has mass M and length 2 m. It is raised by a man on the ferry using a light rope passing a smooth fixed light pulley and connecting to R on the gangplank as shown. R is 1.5 m from end P. Which of the following correctly describes the force required to raise the gangplank steadily?

D

initial force required to raise the gangplank when it is horizontal	*subsequent* force required to raise the gangplank

Initially, take moment about P :

$$T \sin\theta \times d = Mg \times \tfrac{1}{2}L$$

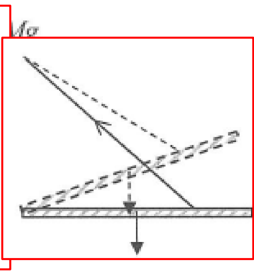
$\therefore \quad T \sin 45° \times 1.5 = Mg \times 1 \qquad \therefore \; T = 0.94\,Mg$

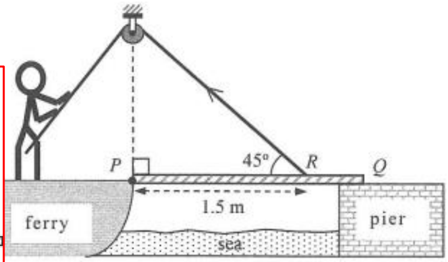
Subsequently, when the plank PQ is pulled up,

the C.G. of the plank would be closer to the end P, i.e., the C.G. would be closer than $\tfrac{1}{2}L$,

$\therefore$ the moment of the weight would decrease, therefore, the tension would decrease.

Note that the angle θ would increase subsequently.





23. Two workmen stand on a wooden plank of negligible weight supported by two trestles as shown.

(a) Find the upward forces exerted on the plank by each the trestle. (3 marks)

(b) Workman Q walks towards X and the plank remains in equilibrium. How will the upward force exerted on the plank by each trestle change? Explain briefly. (5 marks)

(c) How close towards X can Q move without making the plank topple? (2 marks)

(d) When they finish working, P should leave the plank before Q. Explain briefly. (2 marks)

(a) Take moment about X.

Sum of clockwise moment = sum of anticlockwise moment

$$500 \times 3 = F_Y \times 4 + 700 \times 1$$

$$F_Y = 200\text{ N}$$

The upward force acting on the plank by the trestle at Y is 200 N.

Consider the vertical direction.

$F_X + F_Y = 500 + 700$

$F_X = 1200 - 200$

$= 1000\text{ N}$

The upward force acting on the plank by the trestle at X is 1000 N.

(b) Take moment about X.

When Q walks towards X, the clockwise moment produced by him decreases.

Since the plank is in equilibrium, the total anticlockwise moment also decreases.

As the anticlockwise produced by P remains unchanged, the upward force at Y decreases.

Since the total downward force remains unchanged, the total upward force ($F_X + F_Y$) also remains unchanged.

Therefore, the upward force at X increases.

(c) Just before the plank topples, the upward force at Y is zero.

Take moment about X.

Clockwise moment = anticlockwise moment

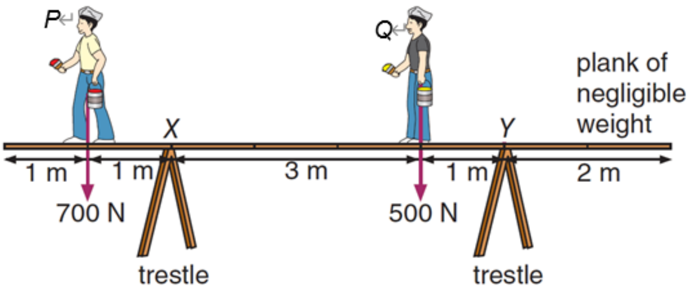
$$500 \times d = 700 \times 1$$

$$d = 1.4\text{ m}$$

Q can move to a position 1.4 m from X before making the plank topple.

(d) If Q leaves the plank before P, there would be no clockwise moment about X

and the plank would topple.



24. A uniform ladder of length 4 m and weight 1000 N leans against a wall making an angle of 60° to the horizontal. The maximum friction between the ladder and the ground is 400 N.

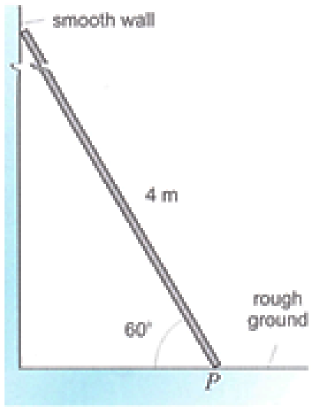


Fig.a

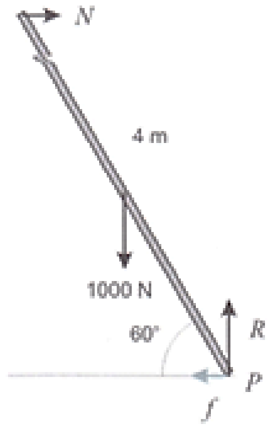


Fig.b

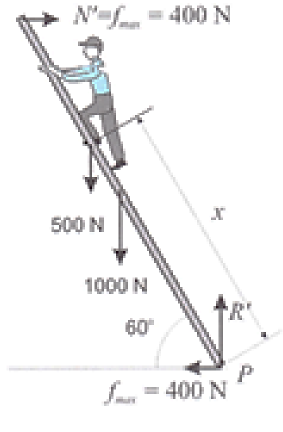


Fig.c

(a) Determine the normal reaction on the ladder from wall.

(b) Hence, find the friction between the ladder and the ground.

(c) A man of weight 500 N climbs up the ladder. How far can the man climb up the ladder without causing the ladder to fall down?

(a) Fig.b shows all the forces acting on the ladder. Taking moment about the lowest point P, the normal reaction N from the wall is given by

$$N \times 4\sin 60^\circ = 1000 \times 2\cos 60^\circ$$

$$\therefore \quad N = \underline{\underline{289 \text{ N}}}$$

(b) The ground exerts two forces on the ladder: the friction f to the left and the normal reaction R upward. As horizontal force are balanced, $f = N = \underline{\underline{289 \text{ N}}}$.

(c) The ladder falls if the friction between the ladder and the ground is not great enough to balance the normal reaction N' from the wall (see Fig.c). The limiting case occurs when

$$N' = f_{max} = 400 \text{ N} .$$

Refer to Fig.c. Taking moment about P,

$$N' \times 4\sin 60^\circ = 1000 \times 2\cos 60^\circ + 500 \times \cos 60^\circ$$

$$400 \times 3.46 = 1000 + 250x$$

$$\therefore \quad x = \underline{\underline{1.54 \text{ m}}}$$